

Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 7: Function Implementation Using MUX.

Objective:

- To get familiarized with **74153[MUX]**;
- To gain experience working with practical circuits.

Required Components:

1. IC 74153
2. IC 7408
3. IC 7432
4. IC 7404

Function:

(i) $F(A, B, C, D) = \sum(1, 3, 4, 7, 8, 9, 11, 12, 14, 15)$

(ii) Build a 3-input OR gate.

Hint. First, find the minterms of a 3-input OR gate system using the truth table below.

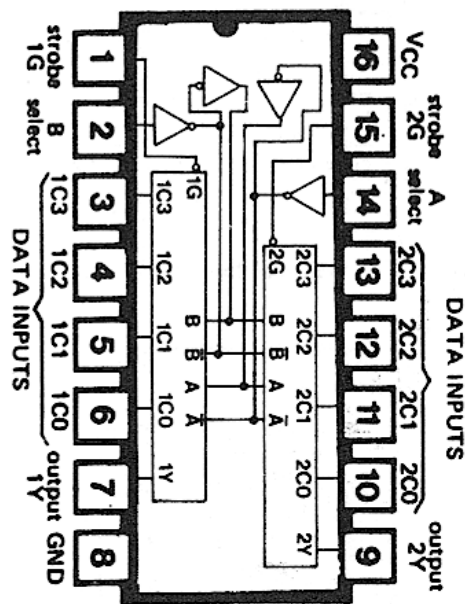
Truth Table of 3-input OR gate:

Input			Output
A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Procedure:

- To design any logic system using a MUX, first identify the minterms of the given system.
- We will use a 4x1 mux to implement the given function; So, first build the table to determine the logic values for the MUX input lines (I0 to I3)
- Remember to connect each IC's VCC pin to the "+5V" position of the DC Power Supply of the trainer board, and the GND or 0V pin to the "GND" position of the trainer board.
- Connect the inputs to the Data switches and outputs to any position of the LED Display.

Pin Diagrams of 74153[MUX] IC:



IC Description:

The IC has two identical 4:1 multiplexers:

- MUX 1: Uses inputs 1C0 to 1C3, output 1Y
- MUX 2: Uses inputs 2C0 to 2C3, output 2Y

Selection Lines: Both MUXs share the same select lines: A and B

Strobe Pins (1G and 2G): Both of these pins are active-low enable pins; So, keep both 1G and 2G pins connected with 0V or “GND”

Report:

Your report must have the following segments:

1. Name of the Experiment.
2. Objective.
3. Required Components.
4. Experimental Setup - Draw the circuit diagrams for both functions.
5. Results - Draw the truth tables of both functions:
6. Discussions - Answer the following questions:
 - a) Design a full adder using 4:1 Mux(s).
 - b) Design a 2-bit Parallel adder using exactly 4 8:1 Mux(s)
 - c) Design a 2-bit Parallel adder cum subtractor using exactly 4 8:1 Mux(s) and if you need any logic gates build that also using Mux. (You can choose any size mux here.)